

PAUL ADIGUN, PhD

PhD in Engineering Mechanics and Energy | University of Tsukuba, Japan

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■ Research Summary

I integrate dynamical meteorology and machine learning to advance the understanding and prediction of climate extremes. My research develops physics-informed neural networks that couple atmospheric dynamics with data-driven methods, with applications to renewable energy resources, precipitation variability, and tropical cyclones.

■ Education

Ph.D. in Engineering Mechanics and Energy

April 2023 – Sep 2026

University of Tsukuba, Japan

Dissertation: *Generalizable Physics-Constrained Machine Learning for Climate Bias Correction Application in Renewable Energy and Extreme Weather Risk Assessment*

M.Sc. in Engineering Mechanics and Energy

April 2021 – March 2023

University of Tsukuba, Japan

Thesis: *Anthropogenic Influence on Extreme Precipitation Over East Asia*

B.Tech in Meteorology

October 2013 – September 2018

Federal University of Technology Akure, Nigeria

Thesis: *Seasonal Rainfall Prediction Using Artificial Neural Network*

■ Academic Experience

Graduate Research Assistant

April – September 2022

University of Tsukuba, Japan

Conducted detection and attribution analyses of anthropogenic climate change signals.

Graduate Teaching Assistant, Mechanics I

April – September 2023

University of Tsukuba, Japan

Assisted in course design, lab supervision, and grading.

Graduate Teaching Assistant, Information Technology

October 2025 – March 2026

University of Tsukuba, Japan

Assisted in course design, lab supervision, and grading.

Graduate Teaching Assistant, Mechanics I

April – September 2026

University of Tsukuba, Japan

Assisted in course design, lab supervision, and grading.

Graduate Research Assistant

April – September 2026

University of Tsukuba, Japan

Developed multivariate bias correction frameworks for CMIP6 climate models; applied ML for tropical cyclone and solar energy simulations; contributed to multiple peer-reviewed publications.

■ Publications

Peer-Reviewed Journal Articles

1. Adigun, P., Dairaku, K., Ogunrinde, A. T. *et al.* Climate-driven synchronization of solar extremes threatens the resilience of Africa's regional power pool. *npj Clean Energy* **2**, 11 (2026). doi:10.1038/s44406-026-00027-7
2. Ogunrinde, A. T., Adigun, P., Dairaku, K. *et al.* Emergent reorganization and increased persistence of meteorological drought across Africa. (Nature) *Communications Earth & Environment* (2026). doi:10.1038/s43247-026-03528-6
3. Ogunrinde, A. T., Adigun, P., Xue, X., Dairaku, K., & Brhane, E. S. (2026). Evaporative Demand Drought Index for Monitoring and Analyzing Drought Conditions in Arid Regions of Asia and Africa. *Weather and Climate Extremes*, 100873. doi:10.1016/j.wace.2026.100873
4. Ogunrinde, A. T., Adigun, P., Xue, X., Dairaku, K., Shameer, S., & Zare, S. (2026). Future wind energy potential for wind farm development in Africa based on bias-corrected NEX-GDDP-CMIP6 wind projections. *Cleaner Engineering and Technology*, 101145. doi:10.1016/j.clet.2026.101145

5. **Adigun, P.**, Dairaku, K., Ogunrinde, A.T., Xue, X., & Ebiendele, P. (2026). Extreme weather and cascading photovoltaic power vulnerabilities under climate change. *Sustainable Energy Technologies and Assessments*, 85, 104776. doi:10.1016/j.seta.2025.104776
6. **Adigun, P.**, Ogunrinde, A.T., Dairaku, K., & Xue, X. (2025). Doubled Saharan dust emissions decrease solar photovoltaic potential through dust–radiation–cloud interactions in CMIP6 simulations. *Environmental Research: Climate*, 100654. doi:10.1088/2752-5295/ae2ef3
7. Ogunrinde, A.T., **Adigun, P.**, Xue, X., Dairaku, K., Shah, S.A., & Adawa, I.S. (2025). Probabilistic quantification of global drought risk amplification from temperature-enhanced evapotranspiration under climate change. *Geoscience Frontiers*, 102235. doi:10.1016/j.gsf.2025.102235
8. **Adigun, P.**, Dairaku, K., & Ebiendele, P. (2025). Physics-constrained deep learning bias correction of CMIP6 solar radiation over Africa and its implications for solar power planning in a changing climate. *Renewable Energy*, 124658. doi:10.1016/j.renene.2025.124658
9. Precious, E., Dairaku, K., & **Adigun, P.A.** (2025). Added value of deep learning downscaling for West African summer monsoon precipitation. *Climate Dynamics*, 63, 410. doi:10.1007/s00382-025-07865-8
10. **Adigun, P.**, Dairaku, K., Ogunrinde, A.T., & Xue, X. (2025). Climate change influence on solar photovoltaic energy production and its associated drivers in CMIP6 ensemble projections. *Journal of Geophysical Research: Atmospheres*, 130, e2024JD042971. doi:10.1029/2024JD042971
11. **Adigun, P.**, Ogunrinde, A.T., Dairaku, K., Adebisi, A.A., & Xue, X. (2025). The future of photovoltaic energy potential in Africa under higher emission scenarios: Insights from CMIP6 multi-model ensemble analysis. *Solar Energy*, 285, 113078. doi:10.1016/j.solener.2024.113078
12. **Adigun, P.**, Abah, E.O., & Ajileye, O.D. (2024). Intensifying human-driven heatwaves characteristics and heat-related mortality over Africa. *Environmental Research: Climate*, 3(1), 015007. doi:10.1088/2752-5295/ad1f41
13. Ogunrinde, A.T., **Adigun, P.**, Xue, X., Dairaku, K., & Qin, J. (2025). Spatiotemporal analysis of drought patterns and trends across Africa: A multi-scale SPEI approach (1960–2018). *International Journal of Digital Earth*, 18(1), 2447342. doi:10.1080/17538947.2024.2447342
14. Ogunrinde, A.T., **Adigun, P.**, Xue, X., Yu, H., Dairaku, K., & Qin, J. (2024). Multi-scale drought variability over West Africa and the associated large-scale circulation patterns. *Geomatics, Natural Hazards & Risk*, 15(1), 2409199. doi:10.1080/19475705.2024.2409199
15. Ogunrinde, A.T., Xian, X., **Adigun, P.**, Yu, H., Dairaku, K., & Qin, J. (2024). Assessing teleconnection influences on the spatial and temporal patterns of meteorological drought in Northwest China. *Big Earth Data*, 8(4), 703–731. doi:10.1080/20964471.2024.2392448
16. **Adigun, P.**, Precious, E.E., Brhane, E.S., et al. (2022). CMIP6 multi-model evaluation of summer extreme precipitation over East Asia. *Modeling Earth Systems and Environment*, 8, 4749–4763. doi:10.1007/s40808-022-01433-3
17. Seun, A.I., **Adigun, P.**, Dairaku, K., & Akande, S.O. (2022). The potential impact of increased urbanization on land surface temperature over South-West Nigeria. *Current Research in Environmental Sustainability*, 4, 100142. doi:10.1016/j.crsust.2022.100142
18. **Adigun, P.** & Ebiendele, E. P. (2019). Seasonal rainfall prediction in Lagos, Nigeria using artificial neural networks. *Asian Journal of Research in Computer Science*, 3(4), 1–10. doi:10.9734/ajrcos/2019/v3i430100
19. Abah, E.O., Kahandage, P.D., Noguchi, R., Ahamed, T., **Adigun, P.**, & Idogho, C. (2025). Assessment of platinum catalyst in rice husk combustion: A comparative life cycle analysis. *MDPI Catalysts*, 15(8), 717. doi:10.3390/catal15080717

Under Review/Submitted

- **Adigun, P.**, Dairaku, K., Ogunrinde, A. T., Precious, E., Nadeem, M. U., Brhane, E. S., & Xue, X. (2026). Historical and projected changes in temperature–precipitation compound hot and dry extremes across Africa based on CMIP5 and CMIP6 ensemble simulations. *EGUsphere [preprint]*. doi:10.5194/egusphere-2026-721
- Ebiendele, P., Adebisi, A. A., Abatzoglou, J., Kramer, R. J., Akinsanola, A. A., **Adigun, P.**, & Chang, I. (2026). Northward Dust-Induced Shift of Tropical Atlantic Precipitation in CMIP6 Models. *EarthArXiv [preprint]*. <https://eartharxiv.org/repository/view/14139/>

Book Chapters

- Abah, E., **Adigun, P.**, Precious, E., Noguchi, R., & Omale, P. (2024). Drought assessment over northern Africa using multi-source satellite product. In T. Ahamed (Ed.), *Remote Sensing Application II* (pp. 379–404). Springer. https://doi.org/10.1007/978-981-97-1188-8_16

Conference Proceedings (Peer-Reviewed)

- **Adigun, P., & Dairaku, K.** (2024). Aerosol Forcing dominating the late-summer precipitation change over East Asia transitional climatic zone in CMIP6 Model Simulation. *Recent Advancements from Aquifers to Skies in Hydrogeology, Geoecology, and Atmospheric Sciences, Proceedings of the 2nd MedGU, Marrakesh 2022 (Volume 1)*, pp. 245–250. link.springer.com/chapter/10.1007/978-3-031-47079-0_55
- **Ebiendele, E. P., Dairaku, K., & Adigun, P.** (2025). Hybrid statistical downscaling in reducing bias in drivers of compound wet–warm extremes during the West African summer monsoon season. *Dynamic Evolution of Atmospheric, Ecological, and Hydrological Systems in Circum-Mediterranean Regions (Advances in Science, Technology & Innovation (ASTI))*, Proceedings of the 2nd MedGU Conference, Marrakesh, 2022) (pp. 117–121). Cham, Switzerland: Springer Nature. link.springer.com/chapter/10.1007/978-3-031-86777-4_22
- **Adigun, P., & Dairaku, K.** (2025). Dynamic-Thermodynamic Coupled Systems Metrics for Improving Tropical Cyclone Intensity Estimation in CMIP6 Models. *Earth Systems and Environment (ESEV-2025) Proceedings*, Springer. **Accepted.**
- **Shobayashi, N., Dairaku, K., & Adigun, P.** (2025). Evaluation of the Influence of Climate Change on Solar Power Potential in Japan. *Earth Systems and Environment (ESEV-2025) Proceedings*, Springer. **Accepted.**
- **Endalew, N. T., Dairaku, K., Adigun, P., & Nguyen, N. K. H.** (2025). Assessing Hydrological Uncertainty in the Upper Blue Nile Basin: A Multi-Model Evaluation. *Earth Systems and Environment (ESEV-2025) Proceedings*, Springer. **Accepted.**
- **Endalew, N. T., Dairaku, K., Nguyen, N. K. H., & Adigun, P.** (2025). Addressing Biases and Enhancing Accuracy: The Role of Downscaled Climate Models in East Africa. *Euro-Mediterranean Journal for Environmental Integration*, Springer. **Accepted.**
- **Nadeem, M. U., Dairaku, K., & Adigun, P.** (2025). Machine Learning-Based Downscaling of High-Resolution GCMs for Accurate Precipitation Tracking in Flood-Prone Regions of Pakistan. *Earth Systems and Environment (ESEV-2025) Proceedings*.
- **Brhane, E. S., Dairaku, K., & Adigun, P.** (2025). Assessing the Hydrological Impacts of Future Land Use Change Scenarios in the Upper Blue Nile Basin Using an Integrated FLUS-SWAT Modeling Approach. *Earth Systems and Environment (ESEV-2025) Proceedings*.
- **Adigun, P., & Dairaku, K.** (2024). Significance of upper troposphere cooling in modulating tropical cyclone potential intensity in the recent decade. *Advances in Science, Technology & Innovation (ASTI)*.
- **Adigun, P., & Dairaku, K.** (2024). Anthropogenic influence on changing extreme precipitation patterns over East Asia. *Advances in Science, Technology & Innovation (ASTI)*.

Oral Presentations

- **Adigun, P., & Dairaku, K.** (2023, November 22). Anthropogenic influence on extreme precipitation over East Asia. *CARE for SEA megacities: Inception workshop and stakeholder consultation*, Bangkok, Thailand.
- **Adigun, P., & Dairaku, K.** (2023, May 23). Dynamic and Thermodynamic component controlling Anthropogenic signal attributed to Extreme precipitation over east Asia in CMIP6 model. *Japan Geoscience Union Meeting 2023 (JpGU)*, Chiba, Japan.
- **Adigun, P., & Dairaku, K.** (2023, February 20). Aerosol Forcing dominating the late-summer precipitation change over East Asia's Transitional Climatic Zone in CMIP6 Model Simulation. *9th International Conference on Flood Management (ICFM9)*.
- **Adigun, P., & Dairaku, K.** (2022, December). Dynamic and Thermodynamic component controlling Anthropogenic signal attributed to Extreme precipitation over east Asia in CMIP6 model simulation. *AGU Fall Meeting Abstracts (Vol. 2022, pp. A41A-03)*.
- **Adigun, P., & Dairaku, K.** (2023, September 27). Greenhouse forcing and its amplification mechanisms on Africa's Hot days and Night. *IITM Hub - International Conference on Regional Climate-CORDEX 2023 (ICRC-CORDEX 2023)*.
- **Adigun, P., & Dairaku, K.** (2023, October 23–27). Detection and attribution of recent trends in temperature extremes over Africa in CanESM5 Large Ensemble. *WCRP Open Science Conference 2023*.
- **Adigun, P., & Dairaku, K.** (2023, October 23–27). Dynamic and Thermodynamic components controlling Anthropogenic signal attributed to Extreme precipitation over east Asia in CMIP6 model simulation. *WCRP Open Science Conference 2023*.
- **Dairaku, K., Kelkar, S., Brhane, E. S., Khan, M. A., Adigun, P., Ebiendele, E. P., & Nguyen, N. K. H.** (2024, June 26). Policy-relevant Regional Climate Information for Societal Needs. *21st Annual Meeting of the Asia Oceania Geosciences Society (AOGS 2024)*, Pyeongchang, Korea.

- Brhane, E. S., Dairaku, K., & **Adigun, P.** (2025, April 28). Assessing the Hydrological Impacts of Future Land Use Change Scenarios in the Upper Blue Nile Basin Using an Integrated FLUS-SWAT Modeling Approach. *1st Earth Systems and Environment Journal Annual Meeting (ESEV-2025)*, Istanbul, Türkiye.
- Shamoto, A., Dairaku, K., **Adigun, P.**, & Defersha, D. T. (2024). Flood Risk Assessment by Atmospheric Rivers. *Japan Geoscience Union Meeting 2024*.
- Dairaku, K., Khan, M. A., Brhane, E. S., Ebiendele, E. P., Nguyen, N. K. H., Llorin, A. G. A., & **Adigun, P.** (2024, October 15). Regional Climate Information for Societal Needs. *Scientific Forum for Numerical Simulation of Asian Regional Climate Change and Earth System Process*, Suzhou, China. [Invited]
- Dairaku, K., Kelkar, S., Brhane, E. S., Khan, M. A., **Adigun, P.**, Tamura, M., & Harada, M. (2022, November 9). Regional climate information for water-energy-health integrated study. *The CORDEX Southeast Asia Workshop*, Bangkok, Thailand.
- Dairaku, K., Kelkar, S., Brhane, E. S., Khan, M. A., **Adigun, P.**, Tamura, M., & Harada, M. (2023, September 26). Regional climate information for water-energy-health integrated study. *International Conference on Regional Climate-CORDEX 2023 (ICRC-CORDEX 2023)*, Pune, India. [Keynote Talk]
- Dairaku, K., Kelkar, S., Brhane, E. S., Khan, M. A., Nguyen, N. K. H., **Adigun, P.**, Tamura, M., & Harada, M. (2023, September 28). Policy-relevant regional climate Information for societal needs. *International Conference on Regional Climate-CORDEX 2023 (ICRC-CORDEX 2023)*, Pune, India. [Invited Talk]
- **Adigun, P.**, Dairaku, K. (2023, September 29). Aerosol Forcing dominating the late-summer precipitation change over East Asia. *International Conference on Regional Climate-CORDEX 2023 (ICRC-CORDEX 2023)*, Pune, India. [Initializing talk]
- Dairaku, K., Ebiendele, E. P., **Adigun, P.**, Brhane, E. S., & Khan, M. A. (2023, November 22). Recent Activities in Asia for Policy-relevant Regional climate information. *CARE for SEA megacities: Inception workshop and stakeholder consultation*, Bangkok, Thailand. [Invited Talk]

Poster Presentations

- Precious, E., Dairaku, K., & **Adigun, P.** “Deep learning downscaling approach for West African compound heatwaves.” *AGU Fall Meeting 2024*, Washington, DC, USA, December 2024.
- **Adigun, P.**, & Dairaku, K. “Dynamic and thermodynamic components controlling anthropogenic signals in extreme precipitation over East Asia.” *AGU Fall Meeting 2022*, Chicago, IL, USA, December 2022.
- Precious, E., Dairaku, K., & **Adigun, P.** “Added value of hybrid bias correction and statistical downscaling in simulating extreme temperatures over Nigeria.” *AGU Fall Meeting 2022*, Chicago, IL, USA, December 16, 2022.
- Precious, E., Dairaku, K., & **Adigun, P.** “Deep Learning Detection of COVID-19, Temporal Variation and its Link with Temperature in Nigeria.” *AGU Fall Meeting 2022*, Chicago, IL, USA, December 13, 2022.
- **Adigun, P.**, Dairaku, K., “Greenhouse forcing and its amplification mechanisms on Africa’s Hot days and Night.” *International Conference on Regional Climate-CORDEX 2023 (ICRC-CORDEX 2023)*, Pune, India, September 29, 2023. [Initializing Talk]
- **Adigun, P.**, Dairaku, K., & Ebiendele, E. P. (2023, May 23). Detection and Attribution of Greenhouse gases and Anthropogenic Aerosols to Extreme Temperatures over Africa in CanESM5 Large Ensemble. *Japan Geoscience Union Meeting 2023 (JpGU)*, Chiba, Japan.

■ Technical Skills

Programming Languages:

- Python (Advanced)
- R (Advanced)
- Fortran (Advanced)
- NCL (Advanced)
- MATLAB (Advanced)

Software:

- QGIS, ArcGIS
- Google Earth Engine
- High-Performance Computing (HPC)

Climate Modeling:

- WRF Model
- RegCM4
- CMIP6 Model Analysis
- Statistical Downscaling
- Bias Correction Methods

Data Analysis Tools:

- CDO/NCO
- FERRET
- GrADS
- Machine Learning (TensorFlow, Scikit-learn)
- Deep Learning (PyTorch, Keras)

■ Certification

2021 Atmospheric Observations of Aerosols Cloud and Reactive Trace Gases. Awarded by Institute for Atmospheric and Earth Research, University of Helsinki Finland

■ Awards and Funding

- **MEXT Scholarship**, Government of Japan (2023-2026)
- **MEXT Scholarship**, Government of Japan (2021-2023)
- **Graduate Climate Conference Travel Grant**, Woods Hole Oceanographic Institution, USA (2024)
- **Habatake Travel Grant**, University of Tsukuba (2024)
- **Department Program Director's Award**, University of Tsukuba (2023)
- **Dean's List Academic Excellent Award**, Federal University of Technology Akure (2017)
- **Northwest Institute of Eco-Environment and Resources Talent Attraction Program** Project titled "Drought Impact Assessment and Predictive Modeling for Arid Regions of Central Asia, West Asia, and North Africa" (Project No. E429020101 2024-2026)
- **Chancellor's Postdoctoral Fellowship**, University of California, Los Angeles (UCLA) — awarded 2026; appointment commencing (Host: Prof. Rong Fu).

■ Professional Service

Peer Reviewer:

- *Climate Dynamics* (Springer)
- *Modeling Earth Systems and Environment* (Springer)
- *Earth Systems and Environment* (Springer)
- *Theoretical and Applied Climatology* (Springer)
- *Atmospheric Research* (Elsevier)
- *Physics and Chemistry of the Earth* (Elsevier)
- *Environmental Research, Climate* (IOP)
- *Environmental Research, Letter* (IOP)

■ Professional Memberships

- American Geophysical Union (AGU)
- Asia Oceania Geosciences Society (AOGS)
- Japan Geoscience Union (JpGU)
- Mediterranean Geosciences Union (MedGU)

■ Languages

- **English:** Native proficiency
- **Japanese:** Conversational

■ Selected Media Coverage

- **"Africa's solar power pools show rising synchronization risk under climate change."** *pv magazine Global*, 8 June 2026. Feature coverage of the *npj Clean Energy* study, with author commentary on reserve margins and the limits of transmission-based diversification.
pv-magazine.com/2026/06/08/africas-solar-power-pools-show-rising-synchronization-risk-under-climate-change
- **"Rising Persistence of Meteorological Drought in Africa."** *Scienmag — Science Magazine*, 25 April 2026. Coverage of the *Communications Earth & Environment* study on the reorganization and increasing persistence of meteorological drought across Africa.
scienmag.com/rising-persistence-of-meteorological-drought-in-africa
- **"Skeptical Science New Research for Week #39 2025."** *Skeptical Science*, 25 September 2025. *Journal of Geophysical Research: Atmospheres* paper on climate change influence on solar photovoltaic energy production featured in the Decarbonization section of the weekly research roundup.
skepticalscience.com/new_research_2025_39.html
- **Daily Briefing.** *Carbon Brief*, UK. By the late 21st century, solar output may drop in tropical areas, due to increased cloud cover and rising temperature, whereas Europe and North America may see improved solar potential.
carbonbrief.org/daily-brief/daily-briefing-1-5c-goal-collapsing-risk-ocean-treaty-into-force-gatwicks-second-runway

■ References

Available upon request.

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